Fractional Dominating Parameters

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Abstract

For an arbitrary subset $\mathcal{P}$ of the reals and a graph G with vertex set V, a function $f\colon V \to \mathcal{P}$ is a $\mathcal{P}$-dominating function of G if the sum of the function values over any closed neighborhood is at least 1. That is, for every $v \in V$, $f(N[v]) \geq 1$. The $\mathcal{P}$-domination number of a graph G is defined to be the infimum of $f(V)$ taken over all $\mathcal{P}$-dominating functions f. When $\mathcal{P} = \{0,1\}$ we obtain the standard domination number, and when $\mathcal{P} = \{-1,0,1\}$ or $\{-1,1\}$ we obtain the minus or signed domination number. In this chapter, we survey some results concerning fractional dominating parameters when $\mathcal{P} = [0,1]$ or $\mathbb{Z}$ or $\mathbb{R}$ in which case we obtain the fractional, integer or real domination numbers, respectively.

Keywords: Fractional domination; Integer domination; Real domination

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1 Introduction

For a graph $G = (V, E)$ with vertex set V and edge set E and for a real-valued function $f\colon V \to \mathbb{R}$, the *weight* of f is $\mathrm{w}(f) = \sum_{v \in V} f(v)$. Further, for $S \subseteq V$ we define $f(S) = \sum_{v \in S} f(v)$; in particular, this means that $\mathrm{w}(f) = f(V)$. For a vertex v in V, the open neighborhood $N(v)$ of v consisting of all neighbors of v; that is, $N(v) = \{u \in V \mid uv \in E\}$. The closed neighborhood $N[v]$ of v consists of v and all neighbors of v; that is, $N[v] = N(v) \cup \{v\}$. For notational convenience, we denote $f(N[v])$ by $f[v]$.

Let $f\colon V \to \{0, 1\}$ be a function which assigns to each vertex of a graph an element of the set $\{0, 1\}$. We say f is a *dominating function* if for every $v \in V$, $f[v] \geq 1$. The function f is a *minimal dominating function* if there does not exist a dominating function $g\colon V \to \{0, 1\}$, $f \neq g$, for which $g(v) \leq f(v)$ for every $v \in V$. This is equivalent to saying that a dominating function f is minimal if for every vertex v such that $f(v) > 0$, there exists a vertex $u \in N[v]$ for which $f[u] = 1$. The domination number and upper domination number of a graph G can be defined as $\gamma(G) = \min\{\mathrm{w}(f) \mid f \text{ is a dominating function on } G\}$ and $\Gamma(G) = \max\{\mathrm{w}(f) \mid f \text{ is a minimal dominating function on } G\}$, respectively.

2 Fractional Domination

We consider here the generalization of domination where vertices have fractional weights in the range $[0, 1]$. A real-valued function $f\colon V \to [0, 1]$ is called a *fractional dominating function* of G if $f[v] \geq 1$ for each $v \in V$. The minimum weight of a fractional dominating function of graph G is the *fractional domination number* $\gamma_f(G)$ of G. Thus, $\gamma_f(G) = \min\{\mathrm{w}(f) \mid f \text{ is a fractional dominating function for } G\}$. While ideas about fractional domination are found in Farber [15] and others, the parameter was formally defined in 1987 by S. T. Hedetniemi reporting on results in [25] at the Eighteenth Southeastern Conference, and in 1988 by Domke, Hedetniemi, and Laskar [12] and in 1990 by Grinstead and Slater [22]. For general information about fractional graph parameters, see [32].

The fractional domination number is readily viewed as a linear program. Thus we can talk of minimum, rather than infimum. The dual of the fractional domination number is the *fractional packing number*. This is the linear programming relaxation of the packing number.

A real-valued function $f\colon V \to [0,1]$ is called a *fractional packing function* of G if $f[v] \le 1$ for each $v \in V$. A fractional packing function f is maximal if there does not exist a fractional packing function $g\colon V \to \{0,1\}$, $f \ne g$, for which $g(v) \ge f(v)$ for every $v \in V$. This is equivalent to saying that a fractional packing function f is maximal if for every vertex v with $f(v) < 1$, there exists a vertex $u \in N[v]$ such that $f[u] = 1$. The maximum weight of a fractional packing function of graph G is the *fractional packing number* $\rho_f(G)$ of G.

Define N as the (closed) neighborhood matrix (that is, the adjacency matrix with the 0's on the diagonal replaced by 1's). We have:

Fractional domination γ_f

minimize $\quad \vec{1}^t\vec{x} = \sum_{i=1}^n x_i$

subject to: $\begin{cases} N\vec{x} \ge \vec{1} \\ x_i \ge 0 \end{cases}$

Fractional packing ρ_f

maximize $\quad \vec{1}^t\vec{y} = \sum_{i=1}^n y_i$

subject to: $\begin{cases} N\vec{y} \le \vec{1} \\ y_i \text{ unrestricted} \end{cases}$

Thus, by the fundamental theorem of linear programming, it follows that:

Theorem 1 *For any graph G, it holds that $\gamma_f(G) = \rho_f(G)$.*

For example, the function f that assigns the weights to the vertices of the prism $G = P_4 \,\square\, K_2$ as illustrated in Figure 1 is a fractional dominating function and a fractional packing function of G of weight $\mathrm{w}(f) = \frac{3}{2}$. Thus by Theorem 1, $\gamma_f(G) = \rho_f(G) = \frac{12}{5}$.

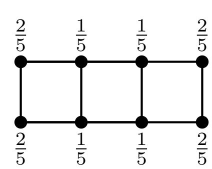

Figure 1. A fractional weighting in a prism $P_4 \,\square\, K_2$

There are immediate bounds involving the minimum and maximum degrees. In particular, weight $1/(\Delta(G)+1)$ on every vertex is a packing function of G and weight $1/(\delta(G)+1)$ on every vertex is a domination function of G. Thus for all graphs, we have the following result first observed by Grinstead and Slater [22] and Domke et al. [12].

Observation 2 ([12, 22]) *If G is a graph of order n with minimum degree $\delta(G) = \delta$ and maximum degree $\Delta(G) = \Delta$, then*

$$\frac{n}{\Delta+1} \leq \rho_f(G) = \gamma_f(G) \leq \frac{n}{\delta+1}.$$

As a consequence of Observation 2 follows the value for a regular graph:

Theorem 3 ([12, 22]) *If G is r-regular graph of order n, then $\gamma_f(G) = \frac{n}{r+1}$.*

Like the ordinary domination number, the parameter $\gamma_f(G)$ ranges between 1 and the order of the graph. Domke et al. [12] showed that $\gamma_f(G) = 1$ if and only if G has a dominating vertex. It is trivial that $\gamma_f(G)$ equals the order if and only if the graph G is empty. It is also trivial that $\gamma_f(G-e) \geq \gamma_f(G)$ where e is any edge in the graph G.

Grinstead and Slater [22] observed an averaging argument: the mean of a collection of dominating functions is itself a dominating function. One can use this argument to "smooth" the function. Thus, for example, one may assume for each orbit of the automorphism group that the weight is the same for all vertices in the orbit. Results on complete bipartite (and complete multipartite graphs) given in [22] and in Domke and Laskar [14] follow:

Theorem 4 ([14, 22]) *For the complete bipartite graph $K_{r,s}$ it holds that*

$$\gamma_f(K_{r,s}) = \frac{r(s-1)+s(r-1)}{rs-1}.$$

The above formula also follows by linear programming duality, since one can readily specify a fractional dominating function and a fractional packing function of the same weight.

For $k \geq 1$ an integer, a function $f\colon V \to \{0, 1, \ldots, k\}$ is called a *k-dominating function* of a graph $G = (V, E)$ if $f[v] \geq k$ for each $v \in V$. The *k-domination* *number*, denoted $\gamma_{\{k\}}(G)$, of G is the minimum weight of a $\{k\}$-dominating function of G. We note that the characteristic function of a dominating set of G is a $\{1\}$-dominating function, and so $\gamma_{\{1\}}(G) = \gamma(G)$. The lower bound with domination can be trivially generalized to $\{k\}$-domination, namely, $\gamma_f(G) \leq \gamma_{\{k\}}(G)/k$. Indeed, Domke et al. [13] showed that

Theorem 5 ([13]) *For any graph G,*

$$\gamma_f(G) = \min_{k \in \mathbb{N}} \left\{ \frac{\gamma_{\{k\}}(G)}{k} \right\}.$$

Domke et al. [11] observed that for a tree, one has equality between the fractional and ordinary domination number. More generally, they proved that for a *block graph* (a graph where every block is complete) equality also holds. This result also follows from more general results proven by Domke et al. [13] involving the packing number.

Theorem 6 ([11]) *If G is a block graph, then $\gamma_f(G) = \gamma(G)$.*

A graph is *chordal* if it contains no cycle of length greater than three as an induced subgraph. A *strongly chordal graph* is a chordal graph that also contains no induced trampoline, where a *trampoline* (or sun) consists of a $2n$-cycle $v_1 v_2 \dots v_{2n} v_1$ in which the vertices v_{2i} of even subscript form a complete graph on n vertices. Iijima and Shibata [27] and Farber [15] showed that for strongly chordal graphs their fractional domination number equals their domination number.:

Theorem 7 ([27, 15]) *If G is a strongly chordal graph, then $\gamma_f(G) = \gamma(G)$.*

This result can be deduced from matrix properties. For, the neighborhood matrix N of a graph is totally balanced if and only the graph is strongly chordal, and linear programs with totally balanced matrices are guaranteed to have integral solutions (see for example [20]).

As was trivially observed, the ordinary domination number is at least the fractional domination number. This suggests the question of how large the domination number can be in terms of the fractional domination number. Chappell, Gimbel and Hartman [5] showed that the ratio can be at most logarithmic in the order:

Theorem 8 ([5]) *If G is a graph, then $\gamma(G) \leq (1 + \ln(1 + \Delta(G)))\gamma_f(G)$.*

This result is essentially best possible as there are r-regular graphs of order n with domination number $\Theta(\frac{n \log r}{r})$; see Alon [1].

2.1 Efficient domination

We say a function $f\colon V \to [0,1]$ is an *efficient fractional dominating function* if for every vertex v it holds that $f[v] = 1$. Equivalently, $N\vec{f} = \vec{1}$ where $\vec{1}$ denotes the all 1's vector in $\mathbb{R}^n$. For example, if G is a regular graph of degree r, then the function f that assigns to each vertex the value $1/(r+1)$ is an efficient fractional dominating function for G. If G is a complete bipartite graph $K_{r,s}$ with partition (R, S), then the function f that assigns to each vertex of R the value $(s-1)/(rs-1)$ and to each vertex of S the value $(r-1)/(rs-1)$ is such a function.

Efficient fractional dominating functions were introduced and first studied in 1988 by Bange, Barkauskas, and Slater [3]. We call an efficient fractional dominating an EFD-*function* (standing for **E**fficient **F**ractional **D**ominating function). Grinstead and Slater [22] called a graph which has an EFD-function "fractionally efficiently dominatable". In 1996, Bange, Barkauskas, Host, and Slater [2] showed that:

Theorem 9 ([2]) *If a graph G is fractionally efficiently dominatable, then every* EFD*-function of G has the same weight.*

We remark that testing for efficiency is a linear program calculation.

If a graph G has an efficient dominating set D, then the characteristic function of D represents a EFD-function. However, the converse is not true: having an efficient fractional dominating function does not imply an efficient dominating set. Consider, for example, any regular graph where n is not a multiple of $r+1$. Another is the graph formed from two 4-cycles by identifying a vertex of each.

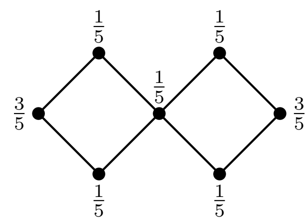

Figure 2. An efficient fractional dominating function in a graph

The converse is true, however, in trees (see [21]), and indeed in block graphs (see [26]):

Theorem 10 ([26]) *If B is a block graph, then B has a* EFD*-function if and only if it has an efficient dominating set.*

Further, note that even if $\gamma_f(G) = \gamma(G)$, it does not necessarily follow that there is an efficient dominating set of G. For example, the graph G shown in Figure 3 satisfies $\gamma_f(G) = \gamma(G) = 3$ and has an EFD-function but does not possess an efficient dominating set.

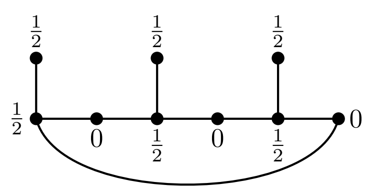

Figure 3. A graph with an EFD-function but no efficient dominating set

2.2 Graph operations

The fractional domination number of the disjoint union of two graphs is just the sum of the individual fractional domination numbers. The result for the join was determined by Fisher [17]: if graphs G and H are not complete, then the optimal fractional dominating set is found by taking optimal fractional dominating sets of both G and H and then scaling all weights of $V(G)$ by some factor and all weights of $V(H)$ by some factor. Optimization yields the result for the join $G+H$

of G and H, which consists of the disjoint union plus edges joining every vertex of G to every vertex of H.

Theorem 11 ([17]) *The fractional domination number of the join of noncomplete graphs G and H is given by*

$$\gamma_f(G+H) = 2 - \frac{\gamma_f(G)+\gamma_f(H)-2}{\gamma_f(G)\gamma_f(H)-1}.$$

Earlier, Fisher, Ryan, Domke, and Majumdar [18] determined the formula for the strong direct product $G \boxtimes H$ of G and H, which has vertex set $V(G) \times V(H)$ and where distinct vertices (u,u') and (v,v') are adjacent in $G \boxtimes H$ if and only if $u = v$ and $u'v' \in E(H)$ or $u' = v'$ and $uv \in E(G)$ or $uv \in E(G)$ and $u'v' \in E(H)$. Their proof rests on the fact that the neighborhood matrix of $G \boxtimes H$ is the tensor product of the neighborhood matrices of G and H.

Theorem 12 ([18]) *For all graphs G and H,*

$$\gamma_f(G \boxtimes H) = \gamma_f(G)\,\gamma_f(H).$$

From this they deduced the bounds for Cartesian product $G \,\square\, H$ of G and H, which has vertex set $V(G) \times V(H)$, and edges between vertices (g,h) and (g',h') if either $gg' \in E(G)$ and $h = h'$, or $g = g'$ and $hh' \in E(H)$.

Theorem 13 ([18]) *For all graphs G and H,*

$$\gamma_f(G \,\square\, H) \geq \gamma_f(G)\,\gamma_f(H).$$

This established the fractional version of Vizing's conjecture. Related results are discussed in [16]. More on the strong product is given by John and Suen [28].

Hare [23] and Stewart and Hare [33] considered the fractional domination numbers of grids with fixed number of rows. Hare showed that the grid $P_2 \,\square\, P_m$ has an efficient dominating set whenever m is odd, and calculated the fractional dominating number for the case that m is even:

Theorem 14 ([23]) *For $m \geq 1$,*

$$\gamma_f(P_2 \,\Box\, P_m) = \begin{cases} \frac{1}{2}(m+1) & \text{if } m \text{ is odd} \\ \frac{m(m+2)}{2(m+1)} & \text{if } m \text{ is even.} \end{cases}$$

The case for $m = 4$ was illustrated in Figure 1. Also Rubalcaba and Walsh [31] noted that every minimal dominating function of $P_2 \,\Box\, P_m$ is a packing function but not every maximal packing function is a dominating function. Values for the product $C_n \,\Box\, P_m$ are given by Xu [34].

2.3 Upper Fractional Domination

A *minimal fractional dominating function* is a fractional dominating function where no value can be lowered. This is equivalent to saying that every vertex u with positive weight has a vertex v in its closed neighborhood such that $f[v] = 1$. The *upper fractional domination number* of G is defined as $\Gamma_f(G) = \max\{\mathrm{w}(f) \mid f$ is a minimal fractional dominating function for $G\}$. The upper parameter $\Gamma_f(G)$ can be expressed as a maximum over a finite collection of linear programs. Hence we write the maximum rather than the supremum. But it is NP-complete in general; see [7]. Upper fractional domination in graphs was introduced and studied by Cheston et al. [7].

The domination and fractional domination parameters are related as follows.

Observation 15 *For every graph G, we have $\gamma_f(G) \leq \gamma(G) \leq \Gamma(G) \leq \Gamma_f(G)$.*

Laskar et al. [30] provided the formula for the upper fractional domination number of the join of two graphs:

Theorem 16 ([30]) *For all graphs G and H,*

$$\Gamma_f(G + H) = \max\{\Gamma_f(G), \Gamma_f(H)\}.$$

Cheston and Fricke [6] established the following result.

Theorem 17 ([6]) *Every (strongly) perfect graph G has $\Gamma_f(G) = \Gamma(G) = \alpha(G)$.*

The convex combination of dominating functions is again a dominating functions. Cockayne, Fricke, Hedetniemi and Mynhardt [9] investigated when the convex combination of minimal dominating functions is again a minimal dominating function.

2.4 Fractional Total Domination

Fractional total domination is defined similarly to fractional domination. A real-valued function $f\colon V \to [0,1]$ in a graph $G = (V, E)$ is called a *fractional total dominating function* of G if the sum of the function values over any open neighborhood is at least 1. That is, for every $v \in V$, $f(N(v)) \geq 1$. The minimum weight of a fractional total dominating function of G is the *fractional total domination number* $\gamma_{ft}(G)$ of G. Thus, $\gamma_{ft}(G) = \min\{\mathrm{w}(f) \mid f$ is a fractional total dominating function for $G\}$. The *upper fractional total domination number* of G is defined as $\Gamma_{ft}(G) = \max\{\mathrm{w}(f) \mid f$ is a minimal fractional total dominating function for $G\}$, where a *minimal fractional total dominating function* is a fractional total dominating function where no value can be lowered.

The characteristic function of a total dominating set is trivially a fractional total dominating function; furthermore, the characteristic function of a minimal total dominating set is a minimal fractional total dominating function. The total domination and fractional total domination parameters are therefore related as follows, where $\Gamma_t(G)$ is the upper total domination number of G.

Observation 18 *For every graph G, $\gamma_{ft}(G) \leq \gamma_t(G) \leq \Gamma_t(G) \leq \Gamma_{ft}(G)$.*

Fricke, Hare, Jacobs and Majumdar [19] established some classes of graphs where the upper total domination and upper fractional total domination numbers are equal.

Theorem 19 ([19]) *If G is a bipartite graph with no induced cycle of length congruent to 2 modulo 4, then $\Gamma_{ft}(G) = \Gamma_t(G)$.*

As with the fractional domination number, the fractional total domination number is readily viewed as a linear program, where here the linear program uses the adjacency matrix A rather than the neighborhood matrix N.

Fractional total domination γ_{ft}	**Fractional open packing** ρ_{fo}
minimize $\quad \vec{1}^t\vec{x} = \sum_{i=1}^n x_i$	maximize $\quad \vec{1}^t\vec{y} = \sum_{i=1}^n y_i$
subject to: $\begin{cases} A\vec{x} \geq \vec{1} \\ x_i \geq 0 \end{cases}$	subject to: $\begin{cases} A\vec{y} \leq \vec{1} \\ y_i \text{ unrestricted} \end{cases}$

Thus, by the fundamental theorem of linear programming, it follows that:

Theorem 20 *For any graph G, it holds that $\gamma_{ft}(G) = \rho_{fo}(G)$.*

For example, if G is an r-regular graph of order n, then the function f that assigns to each vertex the value $1/r$ is a fractional total dominating function and a fractional open packing function of G of weight $\mathrm{w}(f) = n/r$. Thus by Theorem 20, $\gamma_{ft}(G) = \rho_{fo}(G) = n/r$.

Fisher [17] showed that the fractional domination number of a graph and the fractional total domination number of its complement are related as follows.

Theorem 21 ([17]) *For any graph G it holds that*

$$\frac{1}{\gamma_f(G)} + \frac{1}{\gamma_{ft}(\overline{G})} = 1.$$

Proof. If $\gamma_f(G) = 1$, then G has a dominating vertex and so $\overline{G}$ has an isolated vertex. Since such a vertex can have arbitrarily large weight in an open packing, the result follows if we interpret such a graph to have $\gamma_{ft} = \infty$ and $1/\infty = 0$.

So assume $\gamma_f(G) > 1$. Consider a minimum fractional dominating function g of G and a maximum fractional packing h of G. By above we know these both have weight $\gamma_f(G)$. Now for every vertex v, the total weight of g outside v's closed neighborhood $N[v]$ is at most $\gamma_f(G) - 1$. It follows that the function $g/(\gamma_f(G) - 1)$ is a fractional open packing of $\overline{G}$. That is,

$$\gamma_{ft}(\bar{(G)}) \geq \frac{\gamma_f(G)}{\gamma_f(G) - 1}.$$

Similarly, the total weight of h outside $N[v]$ is at least $\gamma_f(G) - 1$, the function $h/(\gamma_f(G) - 1)$ is a fractional total dominating function of $\overline{G}$, and

$$\gamma_{ft}(\bar{(G)}) \leq \frac{\gamma_f(G)}{\gamma_f(G) - 1}.$$

The result follows. $\square$

3 $\mathcal{P}$-domination

In the spirit of fractional domination, Bange, Barkauskas, Host and Slater [2] generalized domination to $\mathcal{P}$-domination for an arbitrary subset $\mathcal{P}$.

Given $\mathcal{P} \subseteq \mathbb{R}$, a function $f\colon V \to \mathcal{P}$ is a $\mathcal{P}$-dominating function of a graph $G = (V, E)$ if the sum of the function values over any closed neighborhood is at least 1. That is, for every $v \in V$, $f[v] \geq 1$. The $\mathcal{P}$-domination number, denoted $\gamma_{\mathcal{P}}(G)$, of G is defined to be the infimum of $f(V)$ taken over all $\mathcal{P}$-dominating functions f.

This concept encompasses several common parameters. When $\mathcal{P} = \{0, 1\}$ we obtain the standard domination number; when $\mathcal{P} = [0, 1]$, we obtain the fractional domination number discussed above; when $\mathcal{P} = \{-1, 0, 1\}$ we obtain the minus domination number, and when $\mathcal{P} = \{-1, 1\}$ we obtain the signed domination number. We remark that discussion of the minus and signed domination numbers can be found in the chapter by Shan and Kang in this book, while discussion of algorithms and complexity of minus and signed domination can be found in the chapter by Hedetniemi, McRae and Mohan in the companion book [**?**].

One important consequence of allowing negative weights, is that for a graph G its $\mathcal{P}$-domination number might be negative. For example, consider a double-star T of order 6 with two vertices of degree 3. If $\mathcal{P} = \mathbb{Z}$ and α is a positive integer, then placing weight α on the two central vertices and weight $1 - \alpha$ on every leaf as illustrated in Figure 4 produced a $\mathcal{P}$-dominating function of total weight $2 - 2\alpha$. One can make α arbitrarily large, implying that $\gamma_{\mathbb{Z}}(T) = -\infty$.

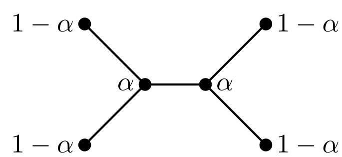

Figure 4. A double-star T with $\gamma_{\mathbb{Z}}(T) = -\infty$

For a subset $\mathcal{P} \subseteq \mathbb{R}$ and a graph $G = (V, E)$, a function $f\colon V \to \mathcal{P}$ is an *efficient $\mathcal{P}$-dominating function* if for every vertex v it holds that $f[v] = 1$. Earlier, Bange et al. [2] when they established Theorem 9 had actually shown:

Theorem 22 *If $\mathcal{P}$ be a subset of the reals $\mathbb{R}$ and if f_1 and f_2 are two arbitrary efficient $\mathcal{P}$-dominating functions for a graph G, then* $\mathrm{w}(f_1) = \mathrm{w}(f_2)$.

A function is *nonnegative* if all the function values are nonnegative. A function which is both nonnegative and efficient $\mathcal{P}$-dominating is called an *NE$\mathcal{P}$D-function* (standing for **N**onnegative **E**fficient $\mathcal{P}$-**D**ominating function). In particular, a NE$\mathbb{R}$D-function (standing for **N**onnegative **E**fficient **R**eal **D**ominating function) in G is a nonnegative efficient real dominating function in G. For example, if $\mathcal{P} = \mathbb{R}$ and G is a regular graph of degree k, then the function f that assigns to each vertex the value $1/(k+1)$ is a NE$\mathbb{R}$D-function for G noting that $f(v) > 0$ and $f[v] = 1$ for every vertex v in the graph G.

In [21], we showed that the question about unbounded weights is directly connected to NE$\mathbb{R}$D-functions.

Theorem 23 ([21]) *For any graph G,*

$$\gamma_{\mathbb{R}}(G) = \begin{cases} \mathrm{w}(f) & \textit{if } G \textit{ has an NE}\mathbb{R}\textit{D-function } f, \\ -\infty & \textit{otherwise.} \end{cases}$$

The proof of Theorem 23 uses linear programming duality and complementary slackness. The dual of real domination is efficient packing. If the dual is feasible, that is, there is an efficient packing function, then that efficient packing function is also an efficient dominating function, and the solution is the size of the efficient dominating function. If the dual is not feasible, then the primal has an unbounded solution.

It follows from Theorem 23 that, for any subset $\mathcal{P}$ of $\mathbb{R}$, if a graph G has an NE$\mathcal{P}$D-function f, then $\gamma_{\mathcal{P}}(G) = \gamma_{\mathbb{R}}(G) = \mathrm{w}(f)$, since f is an NE$\mathbb{R}$D-function of G, and so $\mathrm{w}(f) \geq \gamma_{\mathcal{P}}(G) \geq \gamma_{\mathbb{R}}(G) = \mathrm{w}(f)$.

Another consequence of Proposition 23 is that if $\{0,1\} \subseteq \mathcal{P} \subseteq \mathbb{R}$, and graph G has an efficient dominating set, then $\gamma_{\mathcal{P}}(G) = \gamma(G)$. This follows as the characteristic function of the efficient dominating set is an NE$\mathcal{P}$D-function for G. The converse, however, is not true. For example, the graph G shown in Figure 5 has an NE$\mathcal{P}$D-function f as illustrated, and so $\gamma_f(G) = \mathrm{w}(f) = 3$. Furthermore, it is evident that $\gamma(G) = 3$. However, the graph G does not possess an efficient

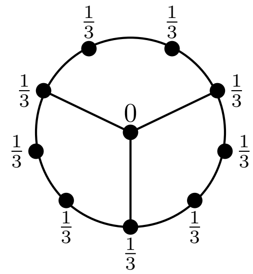

Figure 5. A graph G with no efficient dominating set satisfying $\gamma_{\mathbb{R}}(G) = \gamma(G)$

dominating set. Hence, $\gamma_{\mathbb{R}}(G) = \gamma(G)$ does not necessarily imply that G has an efficient dominating set.

If one takes a $\mathbb{Q}$-dominating function f and multiplies all the weights by the least common multiple of the weights' dominators, one obtains a $\mathbb{Z}$-dominating function. Therefore if there is a $\mathbb{Q}$-dominating function of arbitrarily negative weight, then there is a $\mathbb{Z}$-dominating function too of arbitrarily negative weight. Thus, if $\gamma_{\mathbb{Q}}(G) = -\infty$, then $\gamma_{\mathbb{Z}}(G) = -\infty$.

3.1 Graph classes

By the results of Iijima and Shibata [27] and Farber [15] mentioned earlier, it follows from Theorem 23 that:

Theorem 24 ([15, 21]) *For any strongly chordal graph G,*

$$\gamma_{\mathbb{R}}(G) = \begin{cases} \gamma(G) & \text{if } G \text{ has an NE}\mathbb{R}\text{D-function,} \\ -\infty & \text{otherwise.} \end{cases}$$

For real domination there is no result analogous to Vizing's conjecture. For example, the path P_3 has an efficient dominating set, but the grid $P_3 \,\square\, P_3$ does not have an NE$\mathbb{R}$D-function. Hence, $\gamma_{\mathbb{R}}(P_3 \,\square\, P_3) = -\infty$ while $\gamma_{\mathbb{R}}(P_3) = 1$.

However, Theorem 12 on the strong direct product $G \boxtimes H$ of graphs G and H does generalize. The reason is that, if g and h are dominating functions, then the function $f\colon V(G) \times V(H) \to \mathbb{R}$ defined by $(a, b) \mapsto g(a)h(b)$ (and denoted by $f = g \otimes h$) is a dominating function of $G \boxtimes H$, and if g and h are both efficient then so is f. Furthermore, $w(f) = \mathrm{w}(g) \times \mathrm{w}(h)$. That is,

Theorem 25 *For all graph G and H, for the*

$$\gamma_{\mathbb{R}}(G \boxtimes H) = \begin{cases} \gamma_{\mathbb{R}}(G) \cdot \gamma_{\mathbb{R}}(H), & \textit{if } \gamma_{\mathbb{R}}(G) \textit{ and } \gamma_{\mathbb{R}}(H) \textit{ both positive,} \\ -\infty, & \textit{otherwise.} \end{cases}$$

Brešar, Henning, and Klavzar [4] studied the $\{k\}$-domination number in the Cartesian products of graphs, mostly related to Vizing's conjecture.

Theorem 26 ([4]) *For any graphs G and H,*

$$\gamma_{\{k\}}(G)\gamma_{\{k\}}(H) \le k(k+1)\gamma_{\{k\}}(G \,\square\, H).$$

Theorem 26 simplifies to $\gamma(G)\gamma(H) \le 2\gamma(G \,\square\, H)$ in the special case when $k = 1$, which is the result of Clark and Suen [8].

3.2 Hardness results

If $\mathcal{P} = \mathbb{R}$ or $\mathbb{Q}$, then the determination of $\gamma_{\mathcal{P}}(G)$ can be formulated in terms of solving a linear programming problem, and so can be computed in polynomial-time (see [24] and [29]). On the other hand, the determination of the domination, signed domination and minus domination numbers has been shown to be NP-complete. We [21] showed that the problem is NP-hard provided $\mathcal{P}$ contains 0 and 1 and is bounded from above. On the other hand, there is a linear-time algorithm for finding a minimum $\mathcal{P}$-dominating function in a tree T in many cases.

However, it remains an open problem to determine the complexity of $\mathbb{Z}$-domination. A graph-theoretic proof that shows that $\mathbb{Z}$-domination is a member of NP is not known. This is, however, a consequence of the general result that integer programming is in NP.

4 Other $\mathcal{P}$-parameters

One can define $\mathcal{P}$-analogues of other graphical parameters. In some cases, such as independence number and total domination, one obtains the analogue of Theorem 23; in other cases, such as upper domination, there is an even simpler characterisation of the real version. We discuss here only the independence and upper domination numbers.

Let $G = (V, E)$ be a graph where $|V| = n$ and $|E| = m$. Let I denote the $n \times m$ incidence matrix of G. We say a function $f\colon V \to \mathcal{P}$ is a *$\mathcal{P}$-independence function* of G if for every edge e the sum of the values (weights) assigned under f to the two ends of e is at most 1. The *$\mathcal{P}$-independence number* $\alpha_{\mathcal{P}}(G)$ of G is defined to be the supremum of $w(f)$ taken over all $\mathcal{P}$-independence functions f. If $\mathcal{P} = \{0, 1\}$, then one obtains the ordinary independence number.

An obvious lower bound on $\alpha_{\mathbb{R}}$ is $n/2$ attained by assigning to every vertex a weight of $1/2$. We say a function $g\colon E \to \mathcal{P}$ is an *efficient $\mathcal{P}$-matching function* of a graph $G = (V, E)$ if for every vertex v the sum of the values assigned under g to the edges incident with v is 1. One can obtain a result similar to Proposition 23:

Theorem 27 *For any graph G on n vertices,*

$$\alpha_{\mathbb{R}}(G) = \begin{cases} \frac{1}{2}n, & \text{if } G \text{ has a nonnegative efficient } \mathbb{R}\text{-matching function,} \\ +\infty, & \text{otherwise.} \end{cases}$$

The upper domination number is defined as the cardinality of the largest minimal dominating set of G (see for example, [10]). We say a $\mathcal{P}$-dominating function f is a *minimal $\mathcal{P}$-dominating function* if there does not exist a $\mathcal{P}$-dominating function h, $h \neq f$, such that $h(v) \leq f(v)$ for every $v \in V$. The *upper $\mathcal{P}$-domination number* for G is $\Gamma_{\mathcal{P}}(G) = \sup\{\, f(V) \mid f\colon V \to \mathcal{P} \text{ is a minimal } \mathcal{P}\text{-dominating function on } G\,\}$.

Theorem 28 *Let $\mathcal{P} = \mathbb{Z}$, $\mathbb{Q}$ or $\mathbb{R}$. Then for any connected graph $G = (V, E)$,*

$$\Gamma_{\mathcal{P}}(G) = \begin{cases} 1, & \text{if } G \text{ is complete,} \\ +\infty, & \text{otherwise.} \end{cases}$$

5 Conclusion

In this chapter, we have surveyed some results concerning dominating functions in which negative weights are allowed. We have focused our attention on fractional, integer or real dominating functions in graphs, and discussed the idea of $\mathcal{P}$-dominating functions in graphs, which provides numerous interesting theoretical and computational questions.

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